

Your grade: 90%

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Next item →

☰ Instructions

1.

In general, a smoother decision surface is preferred.

1 / 1 point
- ☒

True

☐

False

✔

Correct

It often indicates better generalization and less overfitting to the training data.

2.

Regularization might cause the performance to go down at the sake of robustness.

1 / 1 point

☒

True

☐

False

✔

Correct

Regularization can reduce overfitting and increase robustness, but it might also lower performance on the training data.

3.

Regularization penalizes weights of an NN layer.

1 point

☒

False

☐

True

✘

Incorrect

Regularization penalizes the weights of a neural network layer to prevent overfitting and improve generalization.

4.

Random Forest model is a good example when weak learners model/capture high level rules.

1 / 1 point

☒

True

☐

False

✔

Correct

Random Forest combines weak decision tree models to capture complex patterns and high-level rules effectively.

5.

Generalization and overfitting work in parallel with each other.

1 / 1 point

☐

True

☒

False

✔

Correct

They are inversely correlated.

6.

Neural network drop-out can help faster optimization.

1 / 1 point

☐

True

☒

False

✔

Correct

It slows down due to NN being pushed to learn alternatives.

7.

In certain situations, bias can be increased to reduce the variance of the error. Despite a higher error, the variability of the error is reduced to achieve a more robust model.

1 / 1 point

☒

True

☐

False

✔

Correct

Increasing bias can reduce variance, leading to a more robust model by stabilizing predictions even if it results in higher overall error.

8.

A linear decision surface is preferred to maximize accuracy.

1 / 1 point

☒

False

☐

True

✔

Correct

Not for the accuracy, but for the robustness and generalization.

9.

Dropout is a neural network optimization algorithm.

1 / 1 point

☐

True

☒

False

✔

Correct

It is a neural network generalization method.

10.

Subsampling features help model abstraction.

1 / 1 point

☒

True

☐

False

✔

Correct

Subsampling features can help model abstraction by reducing complexity and focusing on the most relevant features, leading to better generalization.